Speech in Affective Computing

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Applications (Ayadi et al. 2011)

(1) Man machine interactions on a natural basis
(2) Computer movies and tutorial applications,
(3) Drivers safety via car on—board system in which coded message of the driver's mental state is conveyed to the operating system of car,
(4) Diagnostic tool for a therapist to treat disease
(5) Tool for automatic translation system in which a speaker plays a key role in communication between parties and
(6) Mobile communications

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Difficulties

- Inherent difficulties of deciphering the speaker's emotional state from voice arise due to three factors (Borden et al. 1994)
- ‘what is said’
  - Information of linguistic origin and depends on the way of pronunciation of the words as representatives of the language
- ‘how it is said’
  - Carries paralinguistic information related to the speaker's emotional state
- ‘who says it’
  - Cumulative information regarding the speaker’s basic features and identities like age, gender, and body size

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Types of Database

Three kinds of database (Ververidis and Kotropoulos 2006):

1. Natural
   a. developed on spontaneous speech of real data.
   b. It includes the data recorded from call center conversations, cockpit recordings during abnormal conditions, conversation between a patient and a doctor, conversation with emotions in public places and similar interactions. Batliner et al. (2000).
2. Simulated (Acted)
   a. The speech utterances are collected from experienced, trained and professional artists
3. Elicited (Induced) emotional speech database
   a. Emotions were induced

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Existing Databases

- AIBO Database (Batliner et al. 2004): consists of recording from children while interacting with robot.
  - The database contain 110 dialogues and 29,200 words in 11 emotion categories of anger, boredom, emphatic, helpless, ironic, joyful, reprimanding, rest, surprise and touchy.
- Berlin Database of Emotional Speech (Burkhardt et al. 2005): recordings from 10 actors (5 male, 5 female).
  - The data consist of 10 German sentences recorded in anger, boredom, disgust, fear, happiness, sadness and neutral.
- Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): contains 24 professional actors (12 female, 12 male), vocalizing two statements in a neutral North American accent.

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Existing Databases

- IITKGP-SESC (Koolagudi et al., 2009)
  - The database contains 10 professional artists (5 male and 5 female).
  - For analyzing the emotions they had considered 15 Telugu sentences.
  - Each of the artists had to speak the 15 sentences in 8 basic emotions in one session.
  - The number of sessions considered for preparing the database was 10.
- IITKGP-SEHSC (Rao and Koolagudi 2011)
  - This speech database contains 10 (5 male and 5 female) professional artists for recording purpose collected from All India Radio (AIR) Varanasi, India.
  - The eight emotions considered for recording this database are anger, disgust, fear, happy, neutral, sadness, sarcastic and surprise.
- The total number of utterances recorded in these databases are 12,000 (15 sentences × 8 emotions × 10 artists × 10 sessions)

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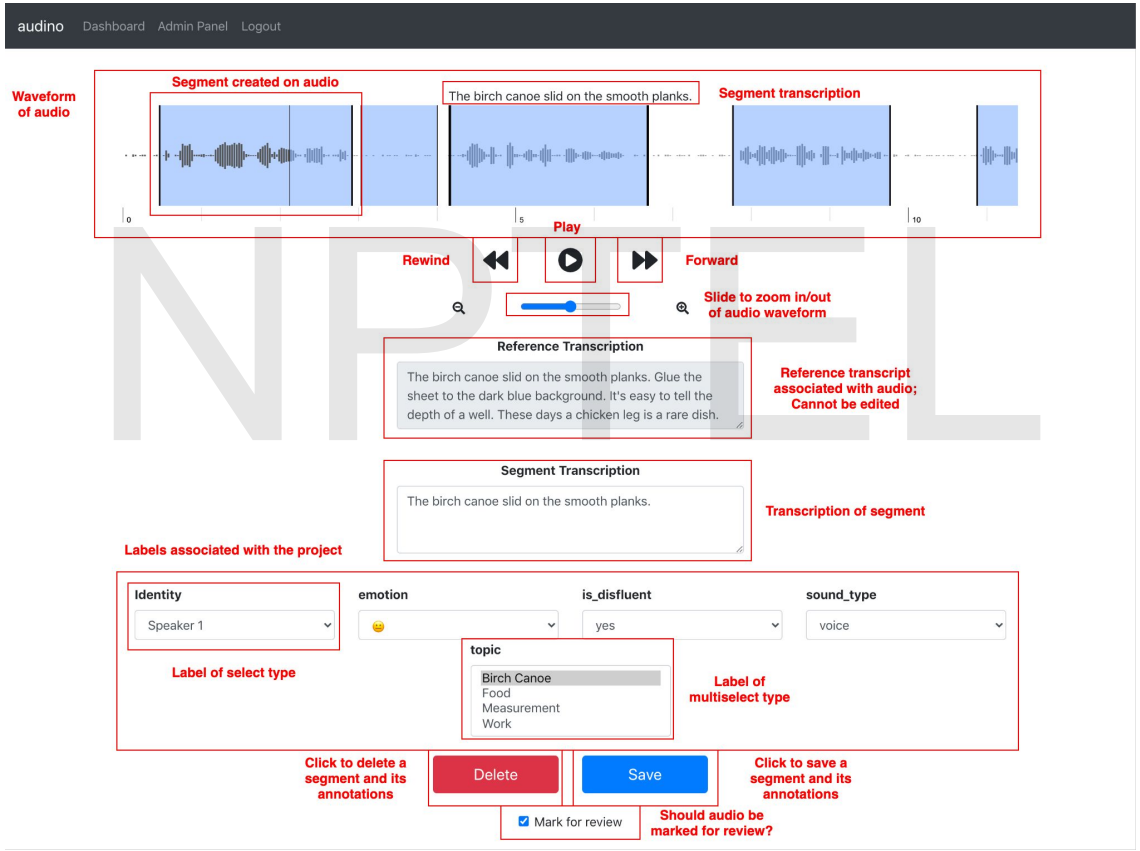
Speech Annotations: Audino



Limitations

- Limited work on non-English languages.
- Limited number of speakers.
- Limited natural databases.
- Limited work on emotion synthesis through speech.
  - https://developer.amazon.com/en-US/blogs/alexa/alexa-skills-kit/2019/11/new-alexa-emotions-and-speaking-styles
- Limited work on cross lingual emotion recognition.
- Insights on ML based methods is available but on DL based methods is limited.

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Acoustic Feature Extraction

- Prosody features: related to rhythm, stress and intonation of speech
  - Fundamental Frequency (fo)
  - Short-term energy
  - Speech-rate; syllable/phoneme rate

- Spectral characteristics measures: related to harmonic or resonant structures
  - Mel frequency cepstral coefficients (MFCCs)
  - Mel filter bank energy coefficients (MFBs)

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Common Prosody Features

- Intensity and amplitude:
  - Loudness: Measures of energy in the acoustic signal
- fo (fundamental frequency)
  - Pitch: Lowest periodic cycle of the acoustic signal
- F1, F2 (formant frequencies):
  - Voice quality: Concentration of acoustic energy around first and second formants
- Speech rate
  - Velocity of speech: Number of complete utterances or elements produced per time unit
- Spectral Energy
  - Timbre: Relative energy in different frequency bands

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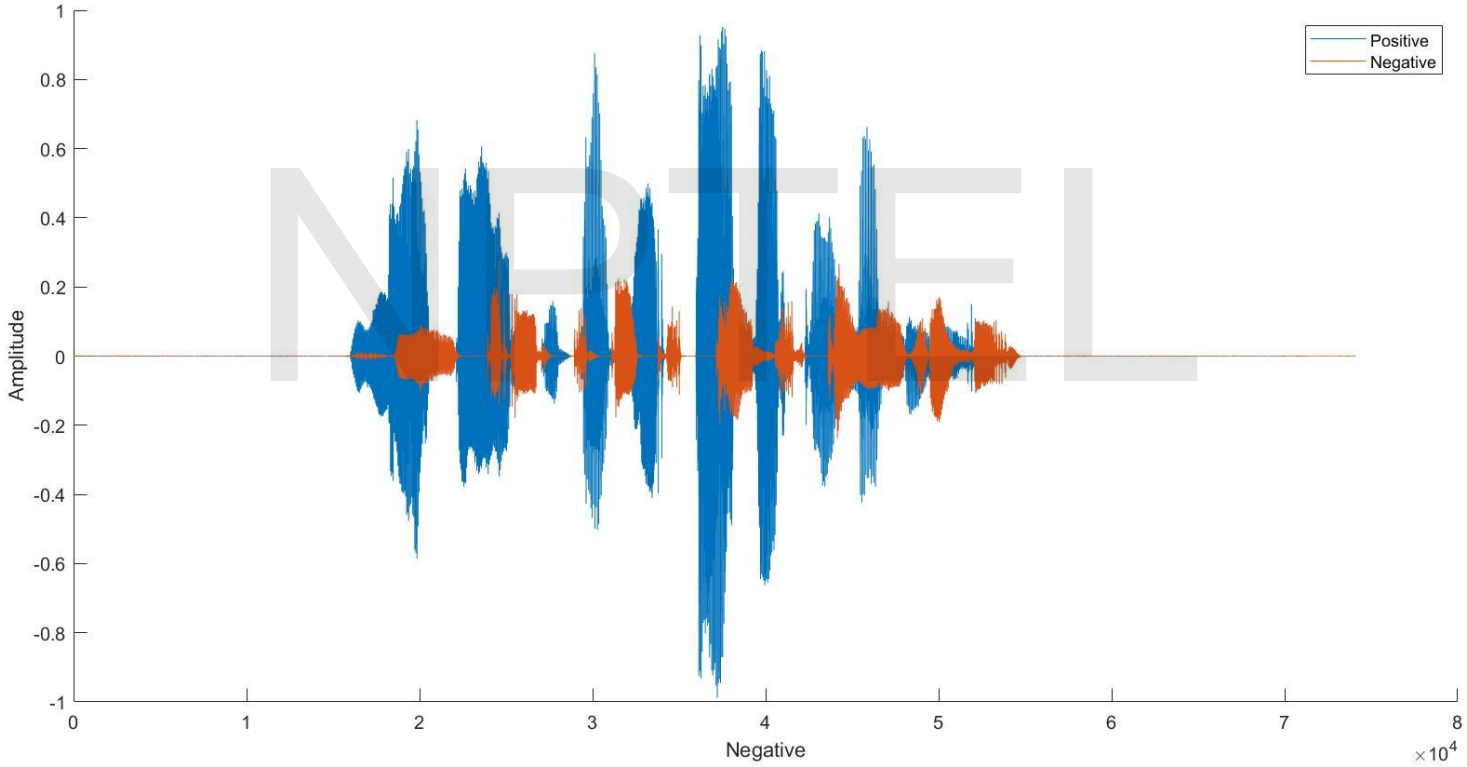
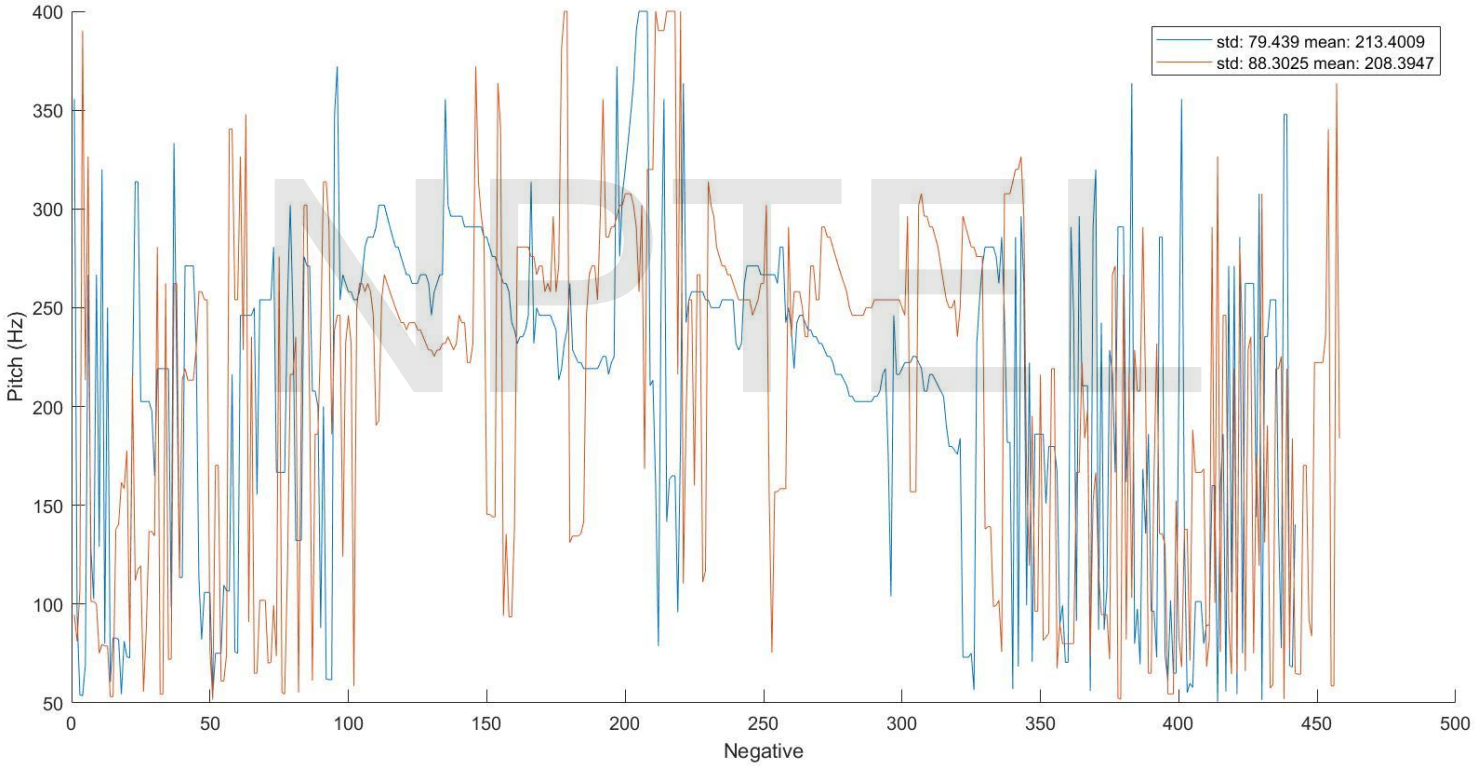
Amplitude: Positive vs Negative



Pitch: Positive vs Negative



Good Vibrations

- Positive voices are generally loud with considerable variability in loudness, have high and variable pitch, and are high in the first two formant frequencies (Kamiloğlu, R. G. et. al.,2020).

- Variations in pitch show differences between high arousal emotions (joy) and low arousal emotions (tenderness and lust), when compared with neutral vocalizations.

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Feature Normalization

1. Angry speech has a higher f_0 values than neutral speech, the emotional differences can be blurred by innerspeaker differences.
2. Speaker normalization to accommodate the differences due to the speakers basic features.
3. E.g. F_0:
   a. Men - 50-250 Hz
   b. Women - 120 - 500 Hz
4. Common Approaches:
   a. Z-score normalization:
      i. Each feature will have zero mean and unit variance across all the data
   b. Min-max Approach
   c. Normal distribution normalization etc.
5. Problem: Can adversely affect the emotional discrimination of the features.

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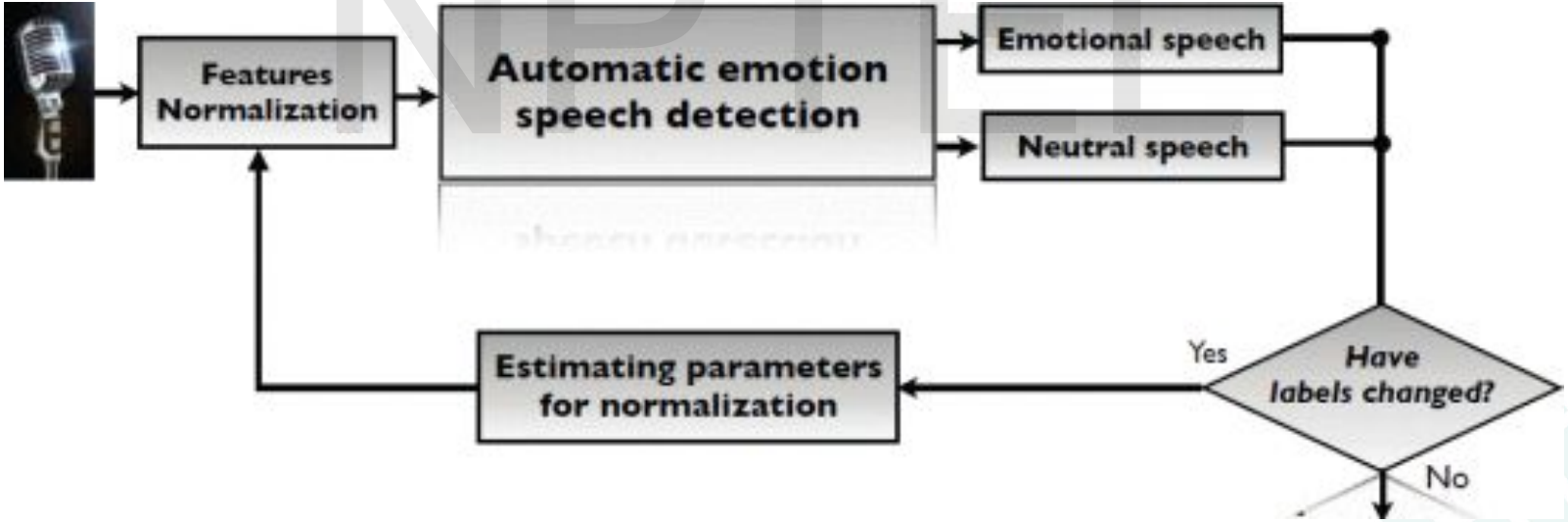
Iterative Feature Normalization

1. Applying a single normalization across the entire corpus can adversely affect the emotional discrimination of the features.
2. Estimating the features using only neutral (non-emotional) samples: IFN (Busso et. al. 2011)



Iterative Feature Normalization

1. First, acoustic features without any normalization are used to detect expressive speech (neutral versus emotional classes).
2. The observations that are labeled as neutral are used to re-estimate the normalization parameters.
3. As the approximation of the normalization parameters improves, the performance of the detection algorithm is expected to improve, leading to better normalization parameters.
4. The process is repeated until the percentage of files in the emotional database that change labels from successive iterations is lower than a given threshold (say 5%).

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Research Challenges

- Interspeech 2009 Emotion Challenge [http://www.compare.openaudio.eu/]
- Audiovisual Emotion Challenge (AVEC 2011) [http://sspnet.eu/avec201x, with x [1–7]]
- Interspeech Computational Paralinguistics challengE
- Emotion in the Wild Challenge
- Multimodal Emotion Challenge (MEC 2016 and MEC 2017) [http://www.chineseldc.org/htdocsEn/emotion.html]

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Open Challenges

- Inter and Intra-speaker variability
  - Heterogenois display of emotions and differences in individuals' vocal structures
  - A speaker can express an emotion in a number of ways and is influenced by the context
- What aspects of emotion production-perception mechanism are captured with the acoustic features?
- Exhaustive and computationally expensive and hence limited real-time applicability

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